

Damián Pablo San Román Alerigi, PhD

Applied Physicist & R&D Lead — Photonics and High-Power Laser Systems

Upstream Advanced Research Center, Saudi Aramco

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Applied physicist and technical lead on Saudi Aramco's high-power laser programme, resident in the Kingdom continuously since 2009 (KAUST, then Saudi Aramco's Upstream Advanced Research Center). Co-led the physics, modelling and field-execution work that carried laser perforation and descaling from laboratory concept to the petroleum industry's **first industrial deployments** — homegrown technology protected by **26 granted patents across eight jurisdictions** (11 pending). Brings international collaborations (University of Southampton hollow-core fibre; KAUST photocatalysis) into industrial application inside Saudi Arabia, and opened research lines in hydrogen-sulfide valorisation and photonic/ML sensing aligned with the Kingdom's energy-diversification objectives.

17 YEARS RESIDENT IN KSA	26 / 11 PATENTS GRANTED / PENDING	8 + ~50 JOURNAL & CONFERENCE PAPERS	437 CITATIONS (GOOGLE SCHOLAR)	21 WORKS AS FIRST / LEAD AUTHOR	2 FIRST INDUSTRIAL DEPLOYMENTS
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CORE EXPERTISE

PHYSICS & ENGINEERING

Laser–matter interaction
Computational electromagnetics
Nonlinear & time-varying media
Hollow-core fibre optics
Photonic sensing

APPLIED / DATA

ML for subsurface inference
Hyperspectral sensing
Computer vision

PROGRAMME LEADERSHIP

Lab-to-field deployment
IP & patent strategy
Academic–industry collaboration

TECHNICAL TOOLKIT

LANGUAGES & TOOLS

MATLAB Python
Fortran C
LaTeX PETSc

SOFTWARE

COMSOL

PROFESSIONAL EXPERIENCE

Applied Physicist — Co-PI, High-Power Laser Programme

2015 – Present · Upstream Advanced Research Center, Saudi Aramco, Dhahran, KSA

Co-led the physics, modelling and field-execution programme that took high-power laser perforation and descaling from laboratory concept to the petroleum industry's **first industrial deployments** (2021–2024), documented across 25+ peer-reviewed and conference publications and a supporting patent portfolio.

- **Physics & modelling foundation (2016–2019):** first-authored the governing numerical treatments of laser–rock interaction and the thermal/geomechanical dynamics of electromagnetic heating; the coupled thermal–mechanical model was published in *SPE Journal* (2019).
- **First-of-kind field deployments (2021–2024):** delivered the industry's first industrial flowline-descaling field trial (Dec. 2021) and first industrial high-power laser perforation deployment (2023), each with a companion field-validation study.
- **Diversification (2020–present):** opened three new research lines — hydrogen-sulfide valorisation to hydrogen, photonic/ML sensing for subsurface and surface assets, and hollow-core optical fibre for kilowatt-class downhole power delivery, developed with the University of Southampton Optoelectronics Research Centre.
- Recognised by Saudi Aramco Senior Vice President Waleed A. Al-Mulhim, with the Laser Research Team, for the first laser-perforation field trial (*The Arabian Sun / Aramco Life*, 2023).

Graduate Researcher, Photonics Laboratory

2009 – 2014 · KAUST, Thuwal, KSA · Advisor: Prof. Boon S. Ooi

- PhD research on light propagation in heterogeneous, nonlinear and time-varying media — computational electromagnetics, beam shaping and chalcogenide-glass material engineering.
- Founding member of the KAUST Photonics Master's programme; developed a high-performance numerical toolbox for wave propagation in complex media (*Maxwell-GEMS*).

Mathematica RSoft

OriginLab

METHODS

FEM FDTD

Spectral / paraxial methods

Inverse-problem design

EDUCATION

PhD, Electrical Engineering (Photonics)

KAUST, Saudi Arabia
2011 – 2014 · Thesis:
Exploring Heterogeneous and Time-Varying Materials for Photonic Applications, Towards Solutions for the Manipulation and Confinement of Light

M.S., Electrical Engineering (Photonics)

KAUST, Saudi Arabia —
Founding cohort
2009 – 2010

Licenciatura (B.S.), Physics

Universidad Nacional Autónoma de México (UNAM)
2003 – 2009 · GPA
9.67/10

IDENTIFIERS

ORCID

0000-0002-3571-6776

SCOPUS AUTHOR ID

55408065500

RESEARCHERID

N-1562-2016

LANGUAGES

Spanish (native) ·
English (fluent) · French
(working)

SELECTED INNOVATIONS (PATENTS)

US 12590836 B2 — Optical energy apparatus, systems, and methods **SOLE INVENTOR**

US 12559366 B2 — Photocatalytic conversion of hydrogen sulfide to hydrogen
LEAD INVENTOR

US 12682216 B2 — Method and system for predicting properties of rock samples using thermal sensing and machine learning **SOLE INVENTOR**

US 12140850 B2 + EP / WO / CN / SA — Reconfigurable optics for beam transformation
LEAD INVENTOR · 4-JURISDICTION FAMILY

US 11725504 B2 + WO / CA / EP / JP — Contactless real-time 3D mapping of surface equipment **LEAD INVENTOR · 5-JURISDICTION FAMILY**

US 11555958 B2 + EP — Nested anti-resonant nodeless optical fiber and subsurface system using the same **WITH UNIVERSITY OF SOUTHAMPTON**

Full record of 26 granted patents and 11 pending applications in the accompanying Extended CV.

SELECTED PUBLICATIONS

Khan *et al.*, “Photochemical valorization of hydrogen sulfide: a study of UV-induced decomposition pathways,” **RSC Advances** 15(44), 37299-37308 (2025)

M. Ferree; J. Kosco; F. Laquai; A. Cavazos Sepulveda; D.P. San Roman Alerigi, “Solar Fuel Production from Hydrogen Sulfide: An Upstream Energy Perspective,” **Adv. Energy Sustain. Res.** (2023) **SENIOR AUTHOR**

Y. Han; Y. Fang; D.P. San Roman Alerigi; S.I. Batarseh, “Thermal/Mechanical Interaction in Laser Perforation Process: Numerical-Model Buildup and Parametric Study,” **SPE Journal** 24(05), 2097-2110 (2019)

D.P. San-Roman-Alerigi; A. Ben Slimane; T.K. Ng; M. Alsunaidi; B.S. Ooi, “A possible approach on optical analogues of gravitational attractors,” **Optics Express** 21(7), 8298-8310 (2013) **FIRST AUTHOR · FEATURED, LASER FOCUS WORLD**

D.P. San-Roman-Alerigi; D.H. Anjum; Y. Zhang; X. Yang; A. Ben Slimane; T.K. Ng; M.N. Hedhili; M. Alsunaidi; B.S. Ooi, “Electron irradiation induced reduction of the permittivity in chalcogenide glass (As₂S₃) thin film,” **J. Appl. Phys.** (2013) **FIRST AUTHOR**

D.P. San-Roman-Alerigi; T.K. Ng; M. Alsunaidi; Y. Zhang; B.S. Ooi, “Generation of J0-Bessel-Gauss beam by an heterogeneous refractive index map,” **JOSA A** (2012) **FIRST AUTHOR**

S.I. Batarseh; D.P. San Roman Alerigi; S. Marshall; R.M. Graves, “First Industrial High-Power Laser Perforation Deployment,” SPE-215016-MS (2023) **KEY MILESTONE**

Full record of 12 peer-reviewed / industry journal articles and ~50 conference papers in the accompanying Extended CV.

RECOGNITION & AFFILIATIONS

- **2013** — Optics Express paper featured in Laser Focus World: Coverage of ‘A possible approach on optical analogues of gravitational attractors’ (Opt. Express 21, 8298).
- **2023** — Saudi Aramco Laser Research Team recognition by SVP Waleed A. Al-Mulhim: Recognition for the first industrial high-power laser perforation field trial.
- **2026** — Gulf Energy Information Excellence Awards — finalist (team/technology recognition): Two technologies named finalists: High-Power Laser Descaler Technology for Surface Flowlines (Best Pipeline Rehabilitation Technology) and WATCH: Wellsite Asset Tracking for Control of Hazards (Best Controls, Instrumentation & Automation Technology – Upstream).
- **2010** — KAUST Academic Excellence Award
- **2009** — KAUST Provost Award
- **2008** — KAUST Discovery Scholar
- **2010** — Professional memberships: IEEE, American Physical Society, Optica/OSA, SIAM, Sociedad Mexicana de Física, The Competitiveness Institute